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To cite this article: Stacey B. Armstrong, Yitong Xin, Nathan D. Sepeda, Martín Polanco, Lynnette A. Averill & Alan K. Davis (2024) Prospective associations of psychedelic treatment for co-occurring alcohol misuse and posttraumatic stress symptoms among United States Special Operations Forces Veterans, *Military Psychology*, 36:2, 184-191, DOI: [10.1080/08995605.2022.2156200](https://doi.org/10.1080/08995605.2022.2156200)

To link to this article: <https://doi.org/10.1080/08995605.2022.2156200>



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Published online: 01 Feb 2023.



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Prospective associations of psychedelic treatment for co-occurring alcohol misuse and posttraumatic stress symptoms among United States Special Operations Forces Veterans

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ABSTRACT

This study evaluated prospective associations of ibogaine and 5-MeO-DMT treatment for risky alcohol use and post-traumatic stress disorder (PTSD) symptoms among United States (US) Special Operations Forces Veterans (SOFV). Data were collected during standard clinical operations at pre-treatment and 1-month (1 m), 3-months (3 m), and 6-months (6 m) post-treatment in an ibogaine and 5-MeO-DMT treatment program in Mexico. Of the 86 SOFV that completed treatment, 45 met criteria for risky alcohol use at pre-treatment (mean age = 44; male = 100%; White = 91%). There was a significant reduction in alcohol use from pre-treatment ($M = 7.2$, $SD = 2.3$) to 1 m ($M = 3.6$; $SD = 3.5$) post-treatment, which remained reduced through 6 m ($M = 4.0$; $SD = 2.9$; $p < .001$, partial eta squared = .617). At 1 m, 24% were abstinent, 33% were non-risky drinking, and 42% were risky drinkers. At 6 m, 16% were abstinent, 31% were non-risky drinking, and 53% were risky drinkers. There were no differences between responders (abstinent/non-risky drinkers) and non-responders (risky drinkers) in demographics/clinical characteristics. However, there were significant and very large differences between responders and non-responders in PTSD symptom ($p < .01$, $d = -3.26$) and cognitive functioning change ($p < .01$, $d = -0.99$). Given these findings, future clinical trials should determine whether psychedelic-assisted therapy holds promise for individuals with complex trauma and alcohol misuse who have not been successfully treated with traditional interventions.

ARTICLE HISTORY

Received 30 June 2022
Accepted 30 November 2022

KEYWORDS

Psychedelic; Special Operations Veterans; Veterans; trauma; alcohol misuse

What is the public significance of this article?— Overall, Special Operations Forces Veterans who sought out psychedelic-assisted therapy experienced a significant and robust reduction in alcohol misuse. Additionally, veterans who experienced a substantial reduction in alcohol misuse had greater reductions in trauma symptoms and improvements in cognitive functioning. Psychedelic-assisted therapy appears to be a hopeful transdiagnostic treatment for this unique population of Veterans, many of whom struggle with various mental health concerns.

Introduction

Special Operations Forces (SOF) is composed of elite personnel chosen because of outstanding physical and psychological resilience who also have advanced training in difficult combat situations (Bartone et al., 2008; Hanwella & de Silva, 2012). However, SOF personnel

are likely to experience many deployments, longer time away from home, more isolation, and potential intense combat situations, which can be traumatic and related to an increased risk of posttraumatic stress disorder (PTSD) and traumatic brain injury (TBI; Hanwella & de Silva, 2012; Hing et al., 2012). To cope with the high pressure of combat, physical injuries, and stress- and trauma- exposure-related psychological disorders, more than 1.5 million Veterans use alcohol which is the most misused substance among military personnel (National Survey on Drug Use and Health, 2013). Alcohol is also commonly used to cope with the effects of TBI (Hanson et al., 2016), a signature injury of conflicts in Iraq and Afghanistan among SOF Veterans (SOFV). Veterans with TBI are more likely to have comorbid psychological and neuropsychiatric issues, including PTSD, alcohol, or drug misuse, and cognitive impairment (Greer et al., 2020). For example, SOFV with combat acquired TBI were two times more likely

to have alcohol- or drug-related diagnoses compared to other Veteran populations (Carlson et al., 2010); and Veterans with a history of moderate-to-severe TBI were more likely to relapse to heavy drinking compared to those without TBI (Jorge et al., 2019).

SOFV have a strong sense of duty and significant mental health-related stigma (Hing et al., 2012), which likely influences their reluctance to seek mental health treatment. Currently, mental health crises and the incidence of suicides among SOFV are increasing at an alarming rate in the context of limited effective treatments for this unique population (Hing et al., 2012; Rocklein Kemplin et al., 2019). More than 20% of high-risk behavior deaths among Veterans were attributed to substance misuse, and about 30% of completed suicides involved drug or alcohol use (National Survey on Drug Use and Health, 2013). Novel treatments are urgently needed to decrease alcohol misuse, stress- and trauma-related symptoms, and suicides in this population. Treatment paradigms must be developed with these complex factors in consideration to increase the health and wellbeing of this population.

Available treatments for traumatic stress and alcohol misuse demonstrate limited efficacy in addressing the complex spectrum of psychiatric symptoms (Watts et al., 2013). Approved psychotherapies for addressing troubling memories (e.g., Cognitive Processing Therapy, Prolonged Exposure, Eye Movement Desensitization, and Reprocessing) and commonly used psychotherapies for addressing alcohol misuse (e.g., cognitive-behavioral therapy, motivational interviewing, mindfulness-based interventions) do not work for all Veterans due to their complex and co-occurring psychiatric symptoms, low response rates for engaged Veterans, and relatively high dropout rates from treatment programs (Bowen et al., 2009; Carroll et al., 2008; Santa Ana et al., 2016; Schnurr et al., 2022; Steenkamp et al., 2015). Further, pharmacotherapies, such as selective serotonin reuptake inhibitors, selective norepinephrine reuptake inhibitors, tricyclic antidepressants, mood stabilizers, antipsychotics, and psychostimulants, are commonly used to treat persistent physiological arousal, mood symptoms, or mitigate cognitive deficits (Puetz et al., 2015). However, they demonstrate limited efficacy for people with PTSD (Krystal et al., 2017), often have undesired side effects and long-term use needs, (Friedman et al., 2007; Lee et al., 2016; Steenkamp et al., 2015) and the two Food and Drug Administration (FDA) approved medications for PTSD (i.e., sertraline, paroxetine) are slow acting and require weeks to take effect. This latency period increases the opportunity for risky behaviors such as self-medication and suicidal ideation/attempt. Pharmacotherapies, such as naltrexone, acamprosate,

and disulfiram, which are commonly prescribed to help reduce alcohol withdrawal symptoms (Rubinsky et al., 2015), exhibit limited efficacy for people with physical (e.g., liver disease) or mental issues and are associated hepatotoxicity, psychosis, and negative body reactions (e.g., headache, fatigue, nausea; Swift & Aston, 2015). Given the comorbid psychological and neuropsychiatric issues among this population, transdiagnostic treatments that can simultaneously address PTSD and commonly overlapping comorbidities, such as alcohol misuse, are urgently needed (Varkovitzky et al., 2018).

Recently, there has been increasing interest in the use of psychedelics as adjuncts to mental health treatment, such as depression, anxiety, substance use disorders (e.g., Bogenschutz et al., 2015; Davis et al., 2020b), and the psychedelic 3,4-methylenedioxy-N-methamphetamine (MDMA) was granted Breakthrough Therapy designation by the FDA in 2017 for the treatment of PTSD. Recent phase 3 results of one clinical trial found that MDMA-assisted therapy was highly efficacious in individuals with severe PTSD, and treatment was safe and well-tolerated, even in those with comorbidities (Mitchell et al., 2021). Furthermore, two recent clinical trials found long-term and robust improvements in alcohol use among participants who were administered psilocybin (Bogenschutz et al., 2015) and MDMA (Sessa et al., 2021). A retrospective study indicated significant and very large self-reported reductions in cognitive impairment and symptoms of PTSD from before to after combined ibogaine- and 5-MeO-DMT-assisted psychedelic therapy among SOFV (Davis et al., 2020a). To address limitations associated with retrospective surveys, a prospective study that followed Veterans for six-month post-treatment was recently completed with a clinical program working with SOFV seeking ibogaine- and 5-MeO-DMT-assisted therapy in Mexico. Study evidence showed that there were significant and large improvements in self-reported PTSD symptoms, post-concussive symptoms, and cognitive functioning from pre-treatment to one-month follow-up which were maintained at six-months (Davis et al., [under review](#)).

Although these preliminary data are encouraging, more research is needed to better understand prospective associations of psychedelic therapy among Veterans who misuse alcohol and experience co-occurring trauma-related symptoms. The goal of the present study was to address this gap in the literature by conducting secondary analysis using a subgroup from the prospective study described above (Davis et al., [under review](#)). The first aim is to provide an overall characterization of alcohol use, PTSD symptoms, and cognitive functioning before and after treatment, through six-month follow-up. The second aim is to explore pre-

treatment and other treatment characteristics that predict response and non-response on alcohol use outcomes.

Method

Clinical program

Data were collected during standard clinical operations at a legal clinical psychedelic program in Mexico between 2019 and 2021. Comprehensive details of the clinical program can be found in a previously published study (Davis et al., 2020a). Treatment involved psychedelic-assisted therapy with ibogaine and 5-MeO-DMT and referral to the program occurred by word of mouth. After being referred, Veterans underwent comprehensive physical and psychological screenings with the program's clinicians to determine participant safety and contraindications.

The clinical residential program lasted three days. Participants underwent screenings (i.e., urine drug screening, alcohol) on the first day to ensure no contraindicated substances were detected. Further, they prepared for their psychedelic-assisted therapy by learning about ibogaine's effects and identified their treatment intentions. Participants whose screenings were clear of any contraindications were administered an oral dose of (10 mg/kg) of ibogaine hydrochloride (99% purity) with up to 5 others in a group setting. Each participant was monitored for cardiovascular and other medical issues while receiving fluids. The next day Veterans were encouraged to journal to process and integrate their experiences. Individual sessions with psychologists and group sessions were also offered to further support integration. On the final day, prior to receiving 5-MeO-DMT, the participants participated in a second preparatory session. At least three inhaled doses (5 mg, 15 mg, and 30 mg) were administered to each participant and additional doses (30 mg, 45 mg) were administered if no desired effects were experienced. The same day, after the effects subsided, Veterans processed and integrated their experiences.

Procedure for clinical program data collection

During initial contact with the treatment program, Veterans were offered the opportunity to participate in this study. After consent and enrolling in the study, participants completed a pre-treatment survey before attending their retreat and then completed follow-up surveys at 1-, 3-, and 6-months after treatment. SurveyGizmo/Alchemer, a secure online survey tool, was used to collect data from Veterans. A staff member not affiliated with the treatment site managed the data collection and no incentives were offered to those who completed surveys.

Evaluable data for the clinical chart review study

Measures administered were part of routine clinical practice at the treatment site (approved by the Institutional Review Board at Ohio State University) to evaluate the efficacy of combined ibogaine- and 5-MeO-DMT-assisted therapy for SOFV. Of the 99 Veterans who participated in treatment during the study period, 86 completed the pre-treatment survey, 71 completed the 1-month follow-up, 62 completed the 3-month follow-up, and 52 completed the 3-month follow-up. 44 participants completed all four surveys. An intention-to-treat model was used to address the missing data, generating a conservative estimate of the treatment effect (Gupta, 2011). Data from the most recently completed time point were carried over if they completed the pre-treatment survey. Thus, the sample was comprised of 86 participants, 45 of which screened positive for alcohol misuse and were included in the current analysis.

Measures

Background characteristics

Veterans were asked items assessing their basic demographics including age, sex, education, military service, ethnicity, marital status, employment status, and state of residence.

Military history

Section A of the National Survey of Veterans (Westat, 2010) was modified to assess military history using 11-items which included military status, military branch, and number of deployments.

Alcohol use

Veterans completed the Alcohol Use Disorders Identification Test-Concise The AUDIT-C which comprises the first three items of the AUDIT (Saunders et al., 1993) and is a validated measure of alcohol use. AUDIT-C scores of ≥ 4 indicated possible misuse (AUDIT-C; Bush et al., 1998). Veterans reported their alcohol consumption before treatment and at each time point after their treatment. The mean of the total AUDIT-C after treatment was subtracted from the total AUDIT-C before treatment to calculate a change score. Cronbach's alphas for the current study ranged from .73 to .88 for each time point.

PTSD symptoms

Veterans completed the Posttraumatic Stress Disorder Checklist (PCL-5) which assesses each of the DSM 5's 21 symptoms of PTSD (Blevins et al., 2015). Veterans were asked to rate how bothered they were (0 = "Not at all" to 4 = "Extremely") by each of the symptoms in the 30 days

before their treatment and at each time point after treatment. The mean of the total PCL-5 after treatment was subtracted from the total PCL-5 from before treatment to calculate a change score. Cronbach's alphas for the current study ranged from .94 to .96 for each time point.

Cognitive symptoms. Veterans completed the Neurobehavioral Symptom Inventory (NSI-22; Cicerone & Kalmar, 1995), a 22-item measure of symptoms commonly associated with post-concussive syndrome and mild traumatic brain injury. Veterans were asked to rate how bothered they were by each symptom on a 5-point scale (0 = "None," 4 = "Very severe"). The mean of the total NSI-22 after treatment was subtracted from the total NSI-22 from before treatment to calculate a change score. Cronbach's alphas for the current study ranged from .92 to .93 for each time point.

Data analysis

Analyses were conducted using IBM SPSS (Version 28). We initially calculated summary statistics of all demographics, background, and treatment history characteristics. To examine aim 1, we ran three separate repeated-measures ANOVAs to provide an overall characterization of alcohol use, PTSD symptoms, and cognitive functioning comparing symptoms from before to after treatment and through six-months follow up. We then calculated change scores for alcohol use, PTSD symptoms, and cognitive functioning by subtracting 1-month follow-up ratings from pre-treatment ratings. To examine aim 2, we ran a series of t-tests to compare the means of responder and non-responder groups at pre-treatment and 1-month follow-up PTSD symptoms, cognitive functioning, and all demographic characteristics to evaluate pre-treatment and other treatment characteristics that predict response and non-response on alcohol use outcomes. Because there were no differences in primary outcome measures beyond 1-month follow-up, we compared pre-treatment to 1-month follow-up scores. Effect sizes for ANOVAs were calculated using the partial eta squared statistic and t-tests were calculated using the Cohen's d statistic and a p-value of .05 was used to determine statistical significance.

Results

Respondent characteristics

Of those that screened positive for possible alcohol misuse at pre-treatment ($N = 45$), 29% screened positive for moderate-risk drinking (AUDIT-C = 4–5), 29% screened positive for high-risk drinking (AUDIT-C = 6–7), and

42% screened positive for severe-risk drinking (AUDIT-C = 8–12). Respondents were largely Caucasian/White (87.2%), non-Hispanic (89.5%), middle-aged ($M = 42.88$, $SD = 7.88$), and male (100%; Table 1).

Change in alcohol misuse, PTSD symptoms, and cognitive functioning

There was a significant and very large reduction in alcohol use from pre-treatment to 1-month post-treatment, which remained reduced through 6 months ($p < .001$, $\eta^2 = .62$). There was also a significant and very large reduction in PTSD symptoms from pre-treatment to 1-month post-treatment, which remained reduced through 6 months ($p < .001$, $\eta^2 = .59$). Lastly, there was a significant and very large improvement in cognitive functioning from pre-treatment to 1-month post-treatment ($p < .001$, $\eta^2 = .52$), which remained reduced through 6 months (Table 2).

Table 1. Demographics of Veterans meeting criteria for risky alcohol use at pre-treatment.

	Veterans with risky alcohol use at pre-treatment ($n = 45$)
	M(SD) or %
Baseline Alcohol Risk	
Moderate	29%
High	29%
Severe	42%
Age	44(7.78)
Sex	
Male	100%
Race	
Caucasian/White	91%
Native American	2%
Mixed Race	7%
Hispanic/Latino	
No	93%
Yes	7%
Education	
High school Diploma or equivalent (GED)	7%
Some college	24%
Trade, technical, or vocational training	4%
Associates degree	2%
Bachelor's degree	31%
Master's degree	29%
Advanced professional degree	2%
Marital or Partnership Status	
Married and living with spouse	58%
Living with partner	2%
Divorced or separated	24%
Never married	16%
Branch of Service	
Army	4%
Navy	87%
Marine Corps	7%
OEF/OIF Service	
Yes	93%
No	7%
Number of Deployments	
1–6	53%
7–12	36%
13–20	11%

Table 2. Change in symptom severity before and after treatment among Veterans meeting criteria for risky alcohol use at pre-treatment.

Measure(N)	Pre-treatment M(SD)	1-month Follow-up M(SD)	3-month Follow-up M(SD)	6-month Follow-up M(SD)	F-test	df	Effect Size	Post Hoc
<i>Alcohol use (45)</i>	7.20(2.31)	3.58(3.53)	3.98(3.21)	3.96(2.92)	22.59***	3	.62	PT>1 m = 3 m = 6 m
<i>PTSD symptoms (19)</i>	37.58(16.43)	20.27(17.80)	18.82(14.21)	18.38(14.82)	20.08***	3	.59	PT>1 m = 3 m = 6 m
<i>Cognitive functioning (45)</i>	32.91(15.22)	20.40(15.48)	21.40(14.16)	20.36(13.83)	14.90***	3	.52	PT>1 m = 3 m = 6 m

* $p < .05$, ** $p < .01$, *** $p < .001$

While, overall, there were significant and very large improvements in risky alcohol use, some Veterans continued to engage in risky drinking. Of the participants who reported risky drinking at pre-treatment, 24% were abstinent, 33% were engaging in non-risky drinking, and 42% were risky drinkers (18% = Moderate risk; 7% = High risk; 18% = Severe risk) at 1-month follow-up. At 3-months, 22% were abstinent, 22% were engaging in non-risky drinking, and 56% were risky drinkers (48% = Moderate risk; 20% = High risk; 32% = Severe risk). At 6-months, 16% were abstinent, 31% were engaging in non-risky drinking, and 53% were risky drinkers (24% = Moderate risk; 16% = High risk; 13% = Severe risk). There were no differences between responders (abstinent/non-risky drinkers; $N = 26$) and non-responders (risky drinkers; $N = 19$) in pre-treatment demographics or pre-treatment clinical characteristics. However, there were significant and very large differences between responders and non-responders in

changes in PTSD symptoms ($p < .01$, $d = -3.26$), and changes in cognitive functioning ($p < .01$, $d = -0.99$; Table 3), from before to 1-month after treatment.

Discussion

This prospective clinical chart review study of combined ibogaine and 5-MeO-DMT treatment among SOFV suggests that this treatment is an effective and robust treatment to reduce problematic alcohol use. Not only were these outcomes established relatively quickly, effects maintained through 6 months post-treatment. There were no demographic differences between treatment responders and non-responders; however, responders had significantly larger reductions in trauma symptoms and improvements in cognitive functioning.

These results are consistent with previous research investigating the impact of psychedelics on alcohol misuse (Garcia-Romeu et al., 2019; Krebs & Johansen,

Table 3. Alcohol use, PTSD symptoms, and cognitive functioning among responders and non-responders.

Measure or demographic variable	Responders @ 1 m ($N = 19$) M(SD)	Non-responders @ 1 m ($N = 26$) M(SD)	t or χ^2	Effect Size
<i>Pre-treatment Alcohol use</i>	7.00(1.94)	7.47(1.78)	-.68	-.20
<i>Pre-treatment PTSD symptoms</i>	38.77(16.48)	35.95(16.67)	.57	.17
<i>PTSD change</i>	-24.73(18.67)	-7.16(16.68)	-3.26**	-.98
<i>Pre-treatment cognitive functioning</i>	31.38(12.84)	35.00(18.15)	-.78	-.24
<i>Cognitive functioning change</i>	-17.69(13.18)	-5.42(11.38)	-3.26**	-.99
<i>Age</i>	43.62(8.12)	44.42(7.48)	-.34	-.10
<i>Race</i>			1.48	
Caucasian/White	53%	38%		
Native American	2%	0%		
Mixed Race	2%	4%		
<i>Hispanic/Latino</i>			.79	
Yes	2%	4%		
No	56%	38%		
<i>Education</i>			6.13	
High school Diploma or equivalent (GED)	2%	4%		
Some college	13%	11%		
Trade, technical, or vocational training	4%	0%		
Associates degree	0%	2%		
Bachelor's degree	22%	9%		
Master's degree	16%	13%		
Advanced professional degree	0%	2%		
<i>Marital or partnership status</i>			1.95	
Married and living with spouse	33%	24%		
Living with partner	0%	2%		
Divorced or separated	13%	11%		
Never married	58%	42%		
<i>OEF/OIF Service</i>			2.35	
Yes	51%	42%		
No	6.7%	0%		
<i>Number of deployments</i>	6.92(4.06)	8.79(5.09)	-.34	-.41

* $p < .05$, ** $p < .01$, *** $p < .001$

2012). Further, studies investigating ibogaine and/or 5-MeO-DMT have shown promise in reducing alcohol use (Davis et al., 2018; Mangini et al., 2021). However, reductions were also related to reductions in trauma symptoms and improvements in cognitive functioning. In terms of trauma, psychedelics, especially MDMA, have been shown to improve PTSD symptoms (Mitchell et al., 2021; Mithoefer et al., 2011). Psychedelics have also shown promise in improving cognitive functioning (Davis et al., 2020a, under review). This research extends the literature by using data from a prospective clinical program evaluation to follow this high-risk population of SOFV who received psychedelic-assisted therapy.

SOFV are particularly high-risk due to their strong sense of duty and significant mental health-related stigma (Hing et al., 2012), and among those with PTSD, dropout rates are often high when treatment is sought (Najavits, 2015). Therefore, the data from this study are particularly beneficial as the treatment is brief and associated with improved symptomology and sustained clinical benefits which is in stark contrast to current therapeutics often used to treat common mental health concerns observed in SOFV.

Some limitations to consider are the program Veterans attended occurred in a naturalistic setting which lacked any controls (e.g., control group, randomization, and blinding) limiting our ability to determine cause and effect relationships. Due to these limitations, we are unable to determine the clinical benefits resulting from each substance or evaluate the benefits of combined psychedelic drug dosing. Furthermore, participation in treatment was determined by self-selection. There may be something meaningful about those who chose to participate in this novel treatment and those who either did not participate or did not participate in survey completion. It is also notable that there were no objective reports, thus increasing risk of self-report bias. Finally, the population evaluated in this study is largely middle-aged, white, and male, which does not reflect the larger Veteran population or the general population of the United States and limits the generalizability of our findings.

Future research should continue to develop and implement well-designed randomized controlled trials (RCTs) that allow for the comparison of ibogaine and 5-MeO-DMT to placebo/control groups. Well-crafted RCTs will allow for longer-term follow-up and assessment of the long-term effects of treatment. Further, researchers should also include validated clinician-administered assessments and survey clinicians and other reporters, as their objective observations of treatment effectiveness is highly valuable. Lastly, because

ongoing stress and continued traumatic experiences are associated with the return to substance use, including measures assessing these variables would be useful.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

Veterans Exploring Treatment Solutions AKD is supported by private philanthropic funding from Tim Ferriss, Matt Mullenweg, Craig Nerenberg, Blake Mycoskie, and the Steven and Alexandra Cohen Foundation. AKD, SBA, NS, and YX are also supported by the Center for Psychedelic Drug Research and Education, funded by anonymous private donors.

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Data availability statement

The data that support the findings of this study are available from the corresponding author, [SBA], upon reasonable request.

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